Yulin (Frank) Zhang

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Education

Duke University, Durham, NC *Expected graduation:* ***May 2028***

B.S. in Computer Science and Mathematics | GPA: **4.0/4.0**

Relevant coursework: Graduate Machine Learning (A+), Graduate Deep Learning (A), Graduate Real Analysis (A)

University of Michigan, Ann Arbor, MI ***2024 - 2025***

B.S. in Computer Science and Mathematics, Honors Program | GPA: **4.0/4.0**

Relevant coursework: Programming & Data Structures (A+), Probability Theory (A+), Honors Multivariable Calculus (A+)

Professional Experience

- **ML Engineer Intern, Baidu** ***Summer 2025***
  - Implemented real-time multi-object detection pipelines in PyTorch and PaddleDetection using YOLOv11 to identify safety-critical items (e.g., hard hats, reflective vests) and human behaviors in industrial video streams.
  - Integrated ByteTrack, BotSort with ReID, etc. to achieve robust multi-object tracking through occlusions and detect wrong-way movements, delivering automated safety compliance alerts and behavior analytics.
- **GenAI Technical Trainer, Scale AI** ***2024 – present***
  - Trained LLMs via RLHF (Reinforcement Learning from Human Feedback) to reduce hallucinations, achieve localization, and align with prompt intent by providing human-guided feedback on multilingual AI responses in culture, math, and coding.
  - Wrote efficient programs that outperform LLM code responses in code generation, reasoning, and refactoring.
  - Optimized response accuracy and safety alignments of generative AI models by rating and editing the generated text.

Research Experience

- **Research Intern, CS+ Program at Duke** ***2026 - present***
  - Studied how LLMs form callable representation of concepts internally to provide a novel mechanistic interpretation of grokking and compression.
- **Research Assistant, Interpretable Machine Learning Lab** ***2026 - present***
  - Designed algorithms for highly interpretable and efficient multiway-split decision trees, utilizing LightGBM for preprocessing and improved SPLIT framework for tree generation, to reduce depth and optimize decision sparsity while improving expressivity, accuracy, and performance.
- **Developer & Research Assistant, DukeNLP Lab** ***2025 – present***
  - Evaluated AI search agents (e.g., Gemini Deep Research) susceptibility to lazy search via search-time contamination by rewriting benchmark questions and evaluating agents' robustness against injected distracting rewritten (Q, A) pairs.
- **Research Assistant, Michigan Institute for Data and AI in Society** ***2024 – 2026***
  - Investigated solutions toward secure Android app use, including detecting or characterizing malicious apps, and suggesting safer alternatives via VirusTotal, Quark-Engine, and custom ML classifiers built on XGBoost, ExcelFormer, and Trompt.
  - Trained an XGBoost model to predict whether an app will be removed from the web store; achieved 80%+ accuracy.
  - Created and validated clean, extensive datasets with VirusTotal and Quark-Engine scan analysis results for comparative study and future research on malware detection.
- **Research Assistant, Living Systems Research Lab at UMich** ***Spring 2025***
  - Researched dynamic cellular movements by animating biological phenomena such as morphogenesis, tissue growth, and pattern formation in MATLAB and Python.
  - Leveraged computational models to simulate complex biophysical processes based on tissue forces and tensions.
- **Research Intern, Columbia University Data Science Institute** ***Summer 2024***
  - Researched algorithms to estimate average casualties in car crashes and efficiently allocate medical resources.
  - Trained robust polynomial regression models in R to predict mean casualties based on driver's age, lighting conditions, etc.; predicted car accidents in the San Francisco area with a minimized mean bias of 0.12 people per accident (RMSE).
- **Research Intern, SoftCom Lab, CS Dept. at Cal State Polytechnic University** ***2020 – 2023***
  - Developed *Concentration*, a Chrome extension that auto-blocks distracting websites based on users' inputs via NLP, LLM, Flask backend, and Firebase database; 2.2K+ users, rated 4.6/5 on Chrome Web Store.
  - Independently worked on *"An NLP-Learning Powered Customizable Approach Towards Auto-Blocking Distracting Websites"* to address digital distractions; filed a provisional patent for the algorithm (app number: 63/383,080)
  - Published and presented a research paper on the NLP algorithm as the lead author at the 9th International Conference on Software Engineering, Copenhagen, Denmark.

Publications

* Denotes equal contribution

- **Zhang, Y.**; Huang, Y.; Chen, S.; Lin, T.; Yang, Z.; Yin, X.; Dhingra, B. *Lazy Grounding: When Search Agents Misapply True Evidence.* (Submitted to EMNLP 2026).

- Chi, B.*; **Zhang, Y.***; Zhong, C.; Rudin, C. *Interpretable Multiway-Split Trees.* (Submitted to NeurIPS 2026).

- **Zhang, Y.**; Sun, Y. *An NLP-Learning Powered Customizable Approach Towards Auto-Blocking Distracting Websites.* https://doi.org/10.5121/csit.2023.130209

- Mohsen, Fadi; **Zhang, Y.**; Bisgin, H.; Buchta, K. *Rebuilding a Balanced Dataset of Removed and Non-Removed Android Apps Using VirusTotal and Quark Features.* (Submitted).

- Bisgin, H.; Buchta, K.; **Zhang, Y.;** Mohsen, Fadi. *Beyond XGBoost: Predicting App Longevity in Google Play with ExcelFormer and Trompt.* (Submitted).

Significant Projects

- **MHacks 2024: MentalAI** ***Sep 2024***
  - Led a team of 4 to fine-tune the GPT-4.0 mini model on datasets of conversations between psychologists and patients to provide mental health services; re-engineered user prompts to maintain the context window.
  - Collaboratively designed front-end UI in React, TypeScript, and Tailwind CSS, with MongoDB database system.
  - Integrated OpenAI API to call the fine-tuned model and optimized real-time responsiveness.

- **Email Automation Service Development** ***2022 – present***
  - Developed an open-source email service to automate timed email delivery for custom recipients; 25K+ downloads.
  - Improved user experience by supporting file attachments, embedded images, custom fonts, etc.
  - Reduced email latency by 30% through maintaining ongoing SMTP (Simple Mail Transfer Protocol) connections.

- **Web Crawler Development for Uni Good News** ***2022 – 2023***
  - Developed a multithreaded web crawler to auto-download 1K+ news articles from the website of Uni Good News, a nonprofit that showcases translated positive news, to provide offline access for visitors in Irvine, CA.
  - Optimized the web crawler to intelligently detect and download new articles on a daily basis.

Awards & Honors

- William J. Branstrom Freshman Prize ***Mar 2025***

  Awarded for maintaining a GPA ranked in the top 5% of College of LSA with all A+ grades across 16 credits in the fall term.

- USACO Gold Division (United States of America Computing Olympiad) ***Dec 2023***

  A national programming contest consisting of challenging, olympiad-level algorithmic problems

  Score: Top 1% out of 12591 participants in Bronze Division; top 10% out of 3841 participants in Silver Division.

Technical Strengths

- Programming languages: Python, R, C++, Java, JavaScript (HTML & CSS), SQL
- Skills: Cybersecurity (*Google Cybersecurity Professional Certificate)*, deep learning frameworks, ML interpretability, backend development